13. Register Write Protection Function

13.1 Overview

The register write protection function protects important registers from being overwritten due to software errors. The registers to be protected are set with the Protect Register (PRCR_S and PRCR_NS).

The two protected registers work on one secure set registers/bits and the other for non-secure set registers/bits. They are collectively mentioned as PRCR.

Table 13.1 lists the association between the PRCR bits and the use of registers to be protected.

The register information to which the PRCR bit is applied is provided in the description of each register.

Table 13.1 Association between PRCR bits and use of registers to be protected

PRCR bit	Register to be protected
PRC0	• Registers related to the clock generation circuit: SCKDIVCR, SCKDIVCR2, SCKSCR, PLLCCR, PLLCR, BCKCR, MOSCCR, HOCOCR, HOCOCR2, MOCOCR, FLLCR1, FLLCR2, CKOCR, OSTDCR, OSTDSR, SOSTDCR, SOSTDSR, PLL2CCR, PLL2CR, PLLCCR2, PLL2CCR2, EBCKOCR, SDCKOCR, SCICKDIVCR, SCICKCR, SPICKDIVCR, SPICKCR, ADCCKDIVCR, ADCCKCR, GPTCKDIVCR, GPTCKCR, LCDCKDIVCR, LCDCKCR, BCKADIVCR, ESWCKDIVCR, ESWPCKDIVCR, ETHPCKDIVCR, BCKACR, ESWCKCR, ESWPCKCR, ETHPCKCR, MOCOUTCR, HOCOUTCR, USBCKDIVCR, OCTACKDIVCR, CANFDCKDIVCR, USB60CKDIVCR, I3CCKDIVCR, USBCKCR, OCTACKCR, CANFDCKCR, USB60CKCR, I3CCKCR, MOSCSCR, HOCOSCR, PLLSCR, PLL2SCR, MOCOSCR, MOSCWTCR, LOCOCR, LOCOUTCR, MOMCR, MOMCR2, SOSCCR, SOMCR, SYRACCR, USBCKSELR
PRC1	• Registers related to the low power modes: OPCCR, PGSCR, PSSTCR0, PSSTCR1, PSSTCR2, PSSTCR3, PSSTCR4, PSSTCR5, PDCTRGD, PDCTRNPU, PDCTRESWM, PDRAMSCR0, PDRAMSCR1, SBYCR, SSCR1, LPSCR, DPSBYCR, DPSWCR, DPSIER0-5, DPSIFR0-5, DPSIEGR0-4, LDOSCR, PLL1LDOCR, PLL2LDOCR, HOCOLDOCR, LVOCR, VSCR, SVSCR, MLSMCR, CPUDSCR, MWMCR • Register related to the battery backup function: VBTBER, VBTICTLR, VBTBKRn (n = 0 to 127), VBTBPCR1, VBTBPCR2, VBTBPSR, VBTADSR, VBTADCR1, VBTADCR2, VBTICTLR2, VBTADCR3, VBTNCWCR
PRC3	• Registers related to the PVD: PVD1CMPCR, PVD2CMPCR, PVD4CMPCR, PVD5CMPCR, PVD1CR0, PVD2CR0, PVD4CR0, PVD5CR0, PVD1CR1, PVD1SR, PVD2CR1, PVD2SR, PVD1FCR, PVD2FCR, PVD4FCR, PVD5FCR, PVDLR, VBATTMNSELR
PRC4	• Registers related to the security and privilege setting registers: CGFSAR, RSTSAR, LPMSAR, PVDSAR, BBFSAR, DPFSAR, DPFSAR1, RSCSAR, PGCSAR, VBRSABAR, VBRPABARS, VBRPABARNS, CPUSAR, DEBUGSAR, CACHESAR, TCMSAR, TCMSABARC, TCMSABARS, IPCSAR, IPCPAR, ICUSARx (x = A, B, E to M), TEVTRCR, BUSSARx (x = A, B, C), BUSPARC, MMPUSARx (x = A, B), DMACCHSAR, DMACCHPAR, DMACSAR, DTCSAR, ELCSARx (x = A, B, C), ELCPARx (x = A, B, C), PmSAR (m = 0 to 9, A, B, C, D), PSARx (x = B, C, D, E), MSSAR, PPARBx (x = B, C, D, E), MSPAR, SRAMSABARn (n = 0 to 3), SRAMSAR, SRAMESAR, MSAR
PRC5	• Registers related to the reset control: SYRSTMSK0-2, TEMPRCR, TEMPRLR

13.2 Register Descriptions

The base address of SYSC is 0x4001_E000 (Secure) and 0x5001_E000 (Non-secure).

Table 13.2 lists the registers related to the register write protection function.

Table 13.2 Register list

Register name	Symbol	Value after reset	Address	Access size
Protect Register for Secure Register	PRCR_S	0x0000	0x4001_E3FA	16
Protect Register for Non-secure Register	PRCR_NS	0x0000	0x5001_E3FE	16

13.2.1 PRCR_S : Protect Register for Secure

Base address: SYSC = 0x4001_E000

Offset address: 0x3FA

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PRKEY[7:0]								—	—	PRC5	PRC4	PRC3	—	PRC1	PRC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRC0	Enable writing to the registers related to the clock generation circuit 0: Write disabled 1: Write enabled	R/W
1	PRC1	Enable writing to the registers related to the low power modes, and the battery backup function 0: Write disabled 1: Write enabled	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	PRC3	Enable writing to the registers related to the PVD 0: Write disabled 1: Write enabled	R/W
4	PRC4	Enables writing to the registers related to the security and privilege setting registers 0: Write disabled 1: Write enabled	R/W
5	PRC5	Enables writing to the registers related to the reset control 0: Write disabled 1: Write enabled	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
15:8	PRKEY[7:0]	PRC Key Code These bits control the write access to the PRCR_S register. To modify the PRCR_S register, write 0xA5 to the upper 8 bits and the target value to the lower 8 bits as a 16-bit unit.	W

Note: S-TYPE6, P-TYPE2

PRCR_S is used to protect registers that are always secure or configured as Secure.

PRC0 bit (Enable writing to the registers related to the clock generation circuit)

The PRC0 bit protects writing to the registers related to the clock generation circuit. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRC1 bit (Enable writing to the registers related to the low power modes, and the battery backup function)

The PRC1 bit protects writing to the registers related to the low power modes and the battery backup function. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRC3 bit (Enable writing to the registers related to the PVD)

The PRC3 bit protects writing to the registers related to the PVD. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRC4 bit (Enables writing to the registers related to the security and privilege setting registers)

The PRC4 bit protects writing to the registers related to the security and privilege setting registers. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

The register controlled by PRC4 may not reflect the PRC4 change when PRCR_S and its controlled registers are continuously written access. Avoid continuous write access or read the PRCR_S after PRC4 change, and then write access the PRC4-controlled register.

PRC5 bit (Enables writing to the registers related to the reset control)

The PRC5 bit protects writing to the registers related to the reset control. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRKEY[7:0]bits (PRC Key Code)

The PRKEY[7:0] bits control permission and prohibition of writing to the PRCR_S register. To write PRCi bits of PRCR_S register, write 0xA5 to the PRKEY[7:0]. In case of writing other than 0xA5 to PRKEY[7:0], PRCi bits do not be changed even if writing to the PRCR_S register.

13.2.2 PRCR_NS : Protect Register for Non-Secure

Base address: SYSC_NS = 0x5001_E000

Offset address: 0x3FE

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PRKEY[7:0]								—	—	—	PRC4	PRC3	—	PRC1	PRC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRC0	Enable writing to the registers related to the clock generation circuit 0: Write disabled 1: Write enabled	R/W
1	PRC1	Enable writing to the registers related to the low power modes, and the battery backup function 0: Write disabled 1: Write enabled	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	PRC3	Enable writing to the registers related to the PVD 0: Write disabled 1: Write enabled	R/W
4	PRC4	Enables writing to the registers related to the privilege setting registers 0: Write disabled 1: Write enabled	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
15:8	PRKEY[7:0]	PRC Key Code These bits control the write access to the PRCR_NS register. To modify the PRCR_NS register, write 0xA5 to the upper 8 bits and the target value to the lower 8 bits as a 16-bit unit.	W

Note: S-TYPE7, P-TYPE2

PRCR_NS is used to protect registers that are configured as Non-secure.

PRC0 bit (Enable writing to the registers related to the clock generation circuit)

The PRC0 bit protects writing to the registers related to the clock generation circuit. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRC1 bit (Enable writing to the registers related to the low power modes, and the battery backup function)

The PRC1 bit protects writing to the registers related to the low power modes and the battery backup function. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRC3 bit (Enable writing to the registers related to the PVD)

The PRC3 bit protects writing to the registers related to the PVD. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

PRC4 bit (Enables writing to the registers related to the privilege setting registers)

The PRC4 bit protects writing to the registers related to the privilege setting registers. Setting this bit to 1 enables writing to the corresponding registers. Setting this bit to 0 disables writing to the corresponding registers.

The register controlled by PRC4 may not reflect the PRC4 change when PRCR_NS and its controlled registers are continuously written access. Avoid continuous write access or read the PRCR_NS after PRC4 change, and then write access the PRC4-controlled register.

PRKEY[7:0]bits (PRC Key Code)

The PRKEY[7:0] bits control permission and prohibition of writing to the PRCR_NS register. To write PRCi bits of PRCR_NS register, write 0xA5 to the PRKEY[7:0]. In case of writing other than 0xA5 to PRKEY[7:0], PRCi bits do not be changed even if writing to the PRCR_NS register.